Chapter 1 — Periodic Table and Periodic Properties

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Historical development and modern periodic law

Q.1 Compare the early attempts at classification of elements with the modern periodic law.

Early chemists tried to arrange elements so that similar properties repeated. Dobereiner grouped elements into triads, Newlands proposed the law of octaves, and Mendeleev arranged elements by increasing atomic mass and left gaps for undiscovered elements. These attempts were useful but incomplete because atomic mass does not always increase in the same order as chemical similarity.

Moseley showed that atomic number, not atomic mass, is the fundamental property of an element. The modern periodic law states that the properties of elements are a periodic function of their atomic numbers. This removes the anomalies of Mendeleev's table and gives a reliable basis for classification.

Attempt	Basis	Main limitation
Dobereiner triads	Groups of three with similar properties	Only a few triads could be identified
Newlands octaves	Every eighth element repeated properties	Failed for heavier elements
Mendeleev table	Increasing atomic mass	Some pairs had to be reversed; isotopes could not be explained
Modern table	Increasing atomic number	Accepted basis of classification

Important Conceptual Short Questions — Topic 1

1. Why is atomic number a more reliable basis of classification than atomic mass?

Atomic number equals the number of protons and fixes the electronic configuration of a neutral atom. Chemical properties depend mainly on valence electrons, so atomic number gives a regular repetition of properties. Atomic mass can vary because of isotopes and does not always follow chemical similarity.

2. How did Moseley’s work improve Mendeleev’s periodic table?

Moseley’s X-ray studies showed that each element has a characteristic nuclear charge, called atomic number. Arranging elements by atomic number removed the need to reverse pairs such as Ar/K and Co/Ni and placed isotopes at the same position.

3. Why are noble gases placed in a separate group?

Noble gases have completely filled valence shells, so they are very unreactive and show little tendency to gain, lose or share electrons. Their distinct behaviour justifies a separate group, Group 18.

Topic 2 — Main features, blocks, families and metallic character

Q.2 Describe the main features of the modern periodic table and explain s, p, d and f blocks.

The modern table has 7 horizontal periods and 18 vertical groups. A period number indicates the principal shell being filled, while elements of a group have the same number of valence electrons and therefore show similar chemical behaviour. Metals occupy the left and centre, non-metals the upper right, and metalloids lie along the stair-step line.

Blocks are named after the subshell that receives the last electron. s-block contains Groups 1 and 2 plus helium; p-block contains Groups 13–18; d-block contains the transition elements; f-block contains the lanthanoids and actinoids. Important families include alkali metals, alkaline earth metals, halogens and noble gases.

Block	Last electron enters	General feature
s	s subshell	Soft, reactive metals; H and He are exceptions
p	p subshell	Metals, metalloids, non-metals and noble gases
d	d subshell	Transition metals; variable oxidation states and coloured ions
f	f subshell	Inner transition metals; placed below the main table

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Important Conceptual Short Questions — Topic 2

1. How can the block of an element be deduced from its electronic configuration?

Look at the subshell occupied by the last electron in the ground-state configuration. If it enters s, p, d or f, the element belongs to that block. For example, an element ending in $3p^5$ is a p-block element.

2. Why do elements in the same group show similar chemical properties?

They have the same number of valence electrons. Since bonding and reactivity mainly involve valence electrons, elements of a group form similar types of compounds and show similar chemical behaviour.

3. How does metallic character change across a period and down a group?

Metallic character decreases across a period because nuclear charge increases and atoms hold electrons more strongly. It increases down a group because atomic size and shielding increase, so outer electrons are lost more easily.

Topic 3 — Periodic arrangement and electronic configuration

Q.3 Explain how electronic configuration determines the position of an element in the periodic table.

The period number is the highest principal quantum number occupied in the ground state. For main-group elements, the group number is related to the number of valence electrons: Groups 1 and 2 have 1 and 2 valence electrons, while Groups 13–18 have 3 to 8 valence electrons. The block is given by the subshell receiving the last electron.

For example, sodium, $1s^2\ 2s^2\ 2p^6\ 3s^1$, is in Period 3, Group 1 and the s-block. Chlorine, ending in $3s^2\ 3p^5$, is in Period 3, Group 17 and the p-block. This connection allows us to predict valency, bonding and reactivity from position.

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Important Conceptual Short Questions — Topic 3

1. Why does the period number correspond to the outermost principal shell?

As atomic number increases across a period, electrons are added to the same principal shell. When a new shell begins, a new period starts. Therefore the highest occupied principal quantum number gives the period.

2. How can valence electrons predict the formula of an ionic compound?

Valence electrons show how many electrons an atom tends to lose, gain or share. For example, a Group 2 metal forms M^{2+} and a Group 17 non-metal forms X^-, so the formula is MX_2.

3. Why do transition metals often show variable oxidation states?

In transition elements, the energies of ns and (n–1)d electrons are close. Different numbers of these electrons can take part in bonding, so more than one oxidation state is possible.

Topic 4 — Atomic and ionic radii

Q.4 Explain the trends in atomic and ionic radii across a period and down a group.

Atomic radius generally decreases across a period because protons are added while electrons enter the same shell. The effective nuclear charge increases and pulls the electron cloud closer. Down a group, atomic radius increases because a new shell is added and inner electrons shield the outer electrons from the nucleus.

A cation is smaller than its parent atom because electron loss reduces repulsion and may remove the outer shell. An anion is larger because added electrons increase repulsion. For isoelectronic ions, radius decreases as nuclear charge increases: $O^{2-} > F^{-} > Na^{+} > Mg^{2+}$.

Species	Trend	Reason
Atom across a period	Radius decreases	Higher effective nuclear charge, same shell
Atom down a group	Radius increases	New shells and shielding
Cation vs atom	Cation smaller	Electron loss; less repulsion
Anion vs atom	Anion larger	Electron gain; more repulsion

Important Conceptual Short Questions — Topic 4

1. Why is Na^{+} smaller than Na?

Sodium loses its single 3s electron to form Na^{+}. The outer shell is removed and the remaining electrons are pulled more strongly by the nucleus, so the ion is smaller.

2. Why is Cl^{-} larger than Cl?

Chlorine gains an electron to form Cl^{-}. The added electron increases electron-electron repulsion in the same shell, so the electron cloud expands.

3. Why is Mg^{2+} smaller than Na^{+} although both are isoelectronic?

Both ions have 10 electrons, but Mg has 12 protons while Na has 11. The greater nuclear charge of Mg pulls the same electron cloud more strongly, giving a smaller radius.

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Topic 5 — Ionization energy

Q.5 Describe the trend in first ionization energy and explain the important exceptions.

First ionization energy is the energy required to remove one electron from one mole of gaseous atoms. It generally increases across a period because atomic size decreases and effective nuclear charge increases. It decreases down a group because the outer electron is farther from the nucleus and more shielded.

Small exceptions occur at Be/B and N/O. Boron has a lower first ionization energy than beryllium because its outer electron is in a higher-energy 2p orbital. Oxygen has a lower first ionization energy than nitrogen because removing one paired 2p electron reduces repulsion, while nitrogen has a relatively stable half-filled 2p subshell.

Important Conceptual Short Questions — Topic 5

1. Why is the second ionization energy of sodium much higher than the first?

The first electron is removed from the outer 3s shell. The second electron would have to be removed from the stable noble-gas core of Na^+, which is much closer to the nucleus and requires far more energy.

2. Why does nitrogen have a higher first ionization energy than oxygen?

Nitrogen has a half-filled 2p subshell, which is relatively stable. In oxygen, one 2p orbital contains a pair of electrons; removing one of these paired electrons reduces repulsion, so it is easier to remove.

3. How does shielding affect ionization energy down a group?

Inner shells repel outer electrons and reduce the nuclear attraction felt by them. Down a group, shielding and distance increase, so the outer electron is removed more easily and ionization energy decreases.

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Topic 6 — Electron affinity and electronegativity

Q.6 Compare electron affinity and electronegativity and explain their periodic trends.

Electron affinity is the energy change when one mole of gaseous atoms gains electrons to form gaseous negative ions. It generally becomes more negative across a period because atoms attract an added electron more strongly. Chlorine has a more negative electron affinity than fluorine because fluorine’s very small 2p shell causes strong electron-electron repulsion.

Electronegativity is the tendency of an atom in a molecule to attract the shared pair of electrons. It increases across a period and decreases down a group. Fluorine is the most electronegative element. Electronegativity difference helps predict whether a bond is non-polar covalent, polar covalent or ionic.

Important Conceptual Short Questions — Topic 6

1. Why is chlorine's electron affinity more negative than fluorine's?

Fluorine is very small, so the added electron enters a compact 2p shell and experiences strong repulsion. Chlorine's larger 3p shell accommodates the electron with less repulsion, so more energy is released.

2. How does electronegativity difference help predict bond polarity?

If two atoms have the same or very similar electronegativity, the shared pair is attracted equally and the bond is non-polar. A moderate difference gives a polar covalent bond, while a large difference favours electron transfer and ionic bonding.

3. Why does electronegativity decrease down a group?

Atomic size and shielding increase down a group. The bonding pair is farther from the nucleus and less strongly attracted, so the atom's ability to attract shared electrons decreases.

Topic 7 — Reactions of sodium and magnesium

Q.7 Compare the reactions of sodium and magnesium with water, oxygen and chlorine.

Sodium is more reactive than magnesium because it loses its outer electron more easily. Sodium reacts vigorously with cold water: $2Na + 2H_2O \rightarrow 2NaOH + H_2$. Magnesium reacts very slowly with cold water but reacts with steam to form magnesium oxide and hydrogen: $Mg + H_2O \rightarrow MgO + H_2$.

Sodium burns in oxygen to give a mixture of sodium oxide and sodium peroxide; under limited oxygen or high temperature, sodium oxide is favoured: $4Na + O_2 \rightarrow 2Na_2O$. Magnesium burns with an intense white flame: $2Mg + O_2 \rightarrow 2MgO$. Both react with chlorine: $2Na + Cl_2 \rightarrow 2NaCl$ and $Mg + Cl_2 \rightarrow MgCl_2$.

Textbook check (p. 14): the printed equation $2Na + O_2 \rightarrow 2Na_2O_2$ is not balanced; the balanced peroxide equation is $2Na + O_2 \rightarrow Na_2O_2$. The printed state MgO(g) should be MgO(s). The steam equation printed as $Mg + 2H_2O \rightarrow MgO + 2H_2$ should be $Mg + H_2O \rightarrow MgO + H_2$.

Important Conceptual Short Questions — Topic 7

1. Why does sodium react more vigorously with water than magnesium?

Sodium has a lower ionization energy and loses its outer electron more easily than magnesium. Therefore it reduces water more rapidly, producing hydrogen gas and heat more vigorously.

2. Why is sodium stored under kerosene oil?

Sodium reacts rapidly with oxygen and moisture in air. Kerosene oil excludes air and water, preventing oxidation and accidental reaction.

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3. How do the reactions of Na and Mg with chlorine show their metallic character?

Both metals lose electrons to chlorine and form ionic chlorides. Sodium loses one electron to form Na^+, while magnesium loses two to form Mg^{2+}, showing their tendency to form positive ions.

Topic 8 — Oxides and chlorides of Period 3

Q.8 Classify Period 3 oxides and chlorides and explain the variation in oxidation number.

Across Period 3, metallic character decreases and non-metallic character increases. Oxides change from basic to amphoteric to acidic. Na_2O and MgO are basic; Al_2O_3 is amphoteric; SiO_2, P_4O_{10}, SO_3 and Cl_2O_7 are acidic. Basic oxides react with acids, acidic oxides react with bases, and amphoteric oxides react with both.

Chlorides change from ionic to covalent. NaCl and $MgCl_2$ are mainly ionic, while $AlCl_3$, $SiCl_4$, PCl_5 and SCl_2 are increasingly covalent. Ionic chlorides dissolve to give nearly neutral solutions, whereas covalent chlorides hydrolyse to give acidic solutions. In the highest oxides, the oxidation number of the Period 3 element rises from +1 in Na_2O to +7 in Cl_2O_7.

Compound type	Example	Nature
Basic oxide	Na_2O, MgO	Reacts with acid
Amphoteric oxide	Al_2O_3	Reacts with acid and base
Acidic oxide	SiO_2, SO_3	Reacts with base
Ionic chloride	NaCl, $MgCl_2$	High melting; neutral solution
Covalent chloride	$SiCl_4$, PCl_5	Hydrolyses; acidic solution

Important Conceptual Short Questions — Topic 8

1. Why does Al_2O_3 behave as an amphoteric oxide?

Aluminium has intermediate metallic and non-metallic character. Its oxide can react with acids as a base and with strong bases as an acid, forming aluminate salts.

2. Why does $SiCl_4$ hydrolyse in water while NaCl does not?

$SiCl_4$ is covalent and silicon can accept lone pairs from water, leading to hydrolysis and HCl formation. NaCl is ionic and simply dissociates into Na^+ and Cl^- without hydrolysis.

3. Why do non-metal oxides generally form acidic solutions with water?

Non-metal oxides are covalent and react with water to form oxoacids, such as H_2SO_4 from SO_3. These acids release H^+ ions, making the solution acidic.

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